This representation, of course, is reducible. Let us compute its character, using the Frobenius formula. We get

$$\begin{aligned}
\chi\begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} &= q(q-1)\nu(x), \\
\chi(A) &= 0 \quad \text{for } A \text{ parabolic or hyperbolic}, \\
\chi\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix} &= \nu\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix} + \nu\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix}^q.
\end{aligned}$$

The last assertion holds because if we regard the matrix as an element of $\mathbb{F}_{q^2}$, conjugation is an automorphism of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$, but the only nontrivial automorphism of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$ is the qth power map.

We thus have

$$\operatorname{Ind}_K^G \mathbb{C}_{\nu^q} \cong \operatorname{Ind}_K^G \mathbb{C}_\nu$$

because they have the same character. Therefore, for $\nu^q \neq \nu$ we get $\frac{1}{2}q(q-1)$ representations.

Next, we look at the tensor product

$$W_1 \otimes V_{\alpha,1},$$

where 1 is the trivial character and W_1 is defined as in the previous section. The character of this representation is

$$\begin{aligned}
\chi\begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} &= q(q+1)\alpha(x), \\
\chi(A) &= 0 \quad \text{for } A \text{ parabolic or elliptic}, \\
\chi\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} &= \alpha(x) + \alpha(y).
\end{aligned}$$

Thus the "virtual representation"

$$W_1 \otimes V_{\alpha,1} - V_{\alpha,1} - \operatorname{Ind}_K^G \mathbb{C}_\nu,$$